

LECTURE 31: MATH1061/7861
TOPICS: COUNTING SELECTIONS
DATE: TUESDAY MAY 12, 2026

Learning objectives:

- (1) Reinforce techniques to solve counting problems.
- (2) Learn the different ways to select r objects from n choices.
- (3) Learn how to count the number of ways in each case.

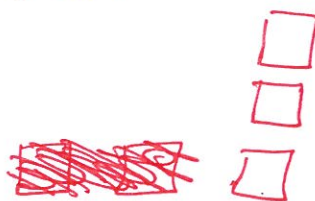
Q1: You are at an ice cream shop and can choose 3 flavours out of 10 to fill your triple scoop cone. How many different possibilities do you have?

Repetition ? Yes or No

Order matters ? Yes or No.

1) Repetition allowed + order matters .

$$10 \times 10 \times 10 = 10^3 = 1000$$



2) Repetition not allowed + order matters

$$\frac{10!}{7!} = 10 \times 9 \times 8 = 720$$

3) Repetition not allowed + order doesn't matter

$$\binom{10}{3} = \frac{10!}{7! 3!} = \frac{10 \times 9 \times 8}{3!} = 120$$

4) Repetition ~~not~~ allowed + order doesn't matter

↳ Exercise : Think about.
(Not amenable).

Q2: There are 10 boys and 11 girls in a classroom. How many ways are there to choose a group of 4 students with exactly 2 girls?

of ways of picking 2 girls:

$$\binom{11}{2} = \frac{11!}{9!2!} = \frac{11 \times 10}{2} = 55$$

of ways of picking boys:

$$\binom{10}{2} = \frac{10 \times 9}{2} = 45.$$

$$\begin{aligned} \underline{\text{Total:}} \quad \binom{11}{2} \times \binom{10}{2} &= 55 \times 45 \\ &= 2475 \end{aligned}$$

41
Q3: There are ~~30~~ people in this classroom.

- (a) What are the total number of possibilities for the number of birthdates (month and day)? Assume no leap years.

Repetition is allowed.

Assume order matters.

$$365^{41}$$

- (b) What is the total number of possibilities ^{if} no two people ~~for~~ have ~~having~~ the same birthdates?

$$\underbrace{365 \times 364 \times \dots \times 325}_{41 \text{ terms}} = \frac{365!}{324!}$$

(c) What is the total number of possibilities ~~for~~ ^{if} at least two people sharing ~~the~~ ^e same birthday?

$$365^{41} - \frac{365!}{324!}$$